# TEMA 10

# Non-Mean-Variance Investment Decisions

The mean-variance investment decision criterion is relevant when asset returns are normally distributed or investors' utility functions are quadratic.[1] In other words, the mean-variance approach holds in the circumstances that the two parameters, mean and variance, are sufficient to describe the investment environment. Therefore, when asset returns are not normally distributed or utility functions are not quadratic, other investment decision tools are needed.[2] This chapter introduces three other portfolio selection criteria: the geometric mean return ( GMR) criterion, the safety-first criterion, and stochastic dominance criterion. Although some of these criteria still use the first and second moment ( mean and variance), they do not require a specific statistical distribution.

## 10.1 GEOMETRIC MEAN RETURN CRITERION

The geometric mean of past historical returns, $\bar{r}_g$, is defined as

$$\begin{aligned}\bar{r}_g &= \left[(1+r_1)(1+r_2)\cdots(1+r_T)\right]^{1/T} - 1\\ &= \left[\prod_{t=1}^{T}(1+r_t)\right]^{1/T} - 1 \qquad (10.1)\end{aligned}$$

Instead of adding together returns to obtain the mean return, one plus the holding period return is serially multiplied. In historical return observations we assume that each of the observations occurs equally likely, that is, with equal probability of $1/T$. Alternatively, when the probability of each return observation is $p_j$, the geometric mean return (GMR) is

$$\bar{r}_g = \prod_{j=1}^{S}(1+r_j)^{p_j} - 1 \qquad (10.2)$$

if we assume $\sum_{j=1}^{S} p_j = 1.0$.

For example, possible returns from an investment will be 15 percent, 5 percent, and $-10$ percent for the next period if the economy is in expansion, normal, and in recession, respectively. The probabilities of the expansion, normal, and recession economy for the next period are 0.3, 0.5, and 0.2, respectively. The GMR is

$$\bar{r}_g = (1 + 0.15)^{0.3}(1 + 0.05)^{0.5}(1 - 0.10)^{0.2} - 1 = 4.63\%$$

### 10.1.1 Maximizing the Terminal Wealth

In a multiperiod setting, it can be shown that the portfolio with the highest expected terminal value is the one with the highest expected GMR. This can readily be seen by noting that

$$\begin{aligned} E\left(W_T\right) &= E\left[W_0\left(1 + r_1\right)\left(1 + r_2\right)\cdots\left(1 + r_T\right)\right] \\ &= W_0\, E\left[\left(1 + \bar{r}_g\right)^T\right] \qquad (10.3) \end{aligned}$$

Because $W_0$ and $T$ are constants, it can be shown that maximizing $E\left(W_T\right)$ is equivalent to maximizing $E$(GMR). Accordingly, the GMR criterion suggests selecting the portfolio at the beginning of each period that has maximum $E$(GMR) in order to maximize $E\left(W_T\right)$. Furthermore, this portfolio will be mean-variance efficient if either (1) returns are lognormally distributed, or (2) returns are normally distributed and investors have logarithmic utility functions.

### 10.1.2 Log Utility and the GMR Criterion

If an investor has a logarithmic utility function, then

$$\begin{aligned} E\left[U\left(W_T\right)\right] &= E\left\{\ln\left[W_0\left(1 + r_1\right)\left(1 + r_2\right)\cdots\left(1 + r_T\right)\right]\right\} \qquad (10.4a) \\ &= \ln W_0 + E\left\{\ln\left[\left(1 + r_1\right)\left(1 + r_2\right)\cdots\left(1 + r_T\right)\right]\right\} \qquad (10.4b) \\ &= \ln W_0 + E\left[\ln\left(1 + \bar{r}_g\right)^{\mathrm{T}}\right] \qquad (10.4c) \\ &= \ln W_0 + T\, E\left[\ln\left(1 + \bar{r}_g\right)\right] \qquad (10.4d) \end{aligned}$$

Because $W_0$ and $T$ are constants, maximizing $E\left[U\left(W_T\right)\right]$ is equivalent to maximizing $E\left[\ln\left(1 + \bar{r}_g\right)\right]$. The portfolio that has the maximum value for $E\left[\ln\left(1 + \bar{r}_g\right)\right]$ will also have the maximum value for $E\left[\left(1 + \bar{r}_g\right)^T\right]$, so, choosing the portfolio with the maximum $E$(GMR) is mathematically equivalent to maximizing both $E\left[U\left(W_T\right)\right]$ and $E\left(W_T\right)$. Thus, the GMR criterion is also an expected utility-based criterion if investors have logarithmic utility functions.

A *myopic* single-period investment rule known as the maximum-expected-log (or MEL) rule has been described by Markowitz (1976), who argues that it is optimal in the long run. This rule is based on the GMR and it states that in each period $t$ the investor should select that portfolio with the maximum expected value of $\ln\left(1 + r_t\right)$. Because equation (10.4b) can be rewritten as

$$E\left[U\left(W_T\right)\right] = \ln W_0 + E\left[\ln\left(1 + r_1\right)\right] + \cdots + E\left[\ln\left(1 + r_T\right)\right] \qquad (10.5)$$

it can be seen that following the MEL rule is equivalent to maximizing $E\left[U\left(W_T\right)\right]$ and is therefore equivalent to maximizing the GMR. Although the MEL rule is an expected utility-maximizing rule if an investor has a logarithmic utility function, it is not an expected utility-maximizing rule for other types of utility functions, even though it maximizes expected terminal wealth. Accordingly, those investors who accept the use of utility theory in decision making reject the MEL rule as a general rule.

An investor who disregards past and future periods in making investment decisions for the current period is said to behave myopically. Using single-period analysis, as in Chapter 17, may thus be described as myopic economic behavior. Unfortunately, the phrase "myopic behavior" in everyday discussions connotes short-sighted behavior that is not good. In contrast, in a portfolio analysis context, delineating and selecting an optimal single-period efficient portfolio will be optimal multiperiod behavior and will maximize multiperiod expected utility under the following three conditions.

1. The investor has positive but diminishing utility of terminal wealth; that is,

$$U'\left(W_T\right) > 0 \text{ and } U''\left(W_T\right) < 0.$$

2. The investor has an isoelastic utility function (which has constant relative risk aversion).[3]
3. The single-period rates of return for securities are distributed according to a multivariate normal distribution and are independent of prior period returns.

These conditions simplify the mathematics sufficiently to prove that expected utility in the multiperiod case can be maximized by acting myopically.[4] For example, if the investor has a logarithmic utility function $U = \ln W_T$, where $W_T = W_0\left(1+r_1\right)\left(1+r_2\right)\cdots\left(1+r_T\right)$, then for any time $t$ between 0 and $T$, the investor will act to maximize the derived utility function, which in this case is equivalent to maximizing $E\left[U\left(W_T\right)\right]$ or $E\left[\ln\left(1+r_t\right)\right]$.

### 10.1.3 Diversification and the GMR

The previous section showed that selecting the portfolio with the greatest GMR is equivalent to maximizing the expected terminal wealth, $E\left(W_T\right)$, and, also, the expected utility, $E\left[U\left(W_T\right)\right]$, when the utility function is logarithmic. Another good feature of the geometric mean criterion in investment decision making is that the portfolio with the greatest GMR is usually a diversified portfolio. Suppose there are three stocks whose rates of return over five periods are given in Table 10.1. Among these three stocks, stock B provides the largest terminal wealth in five periods because it has the greatest GMR. Specifically, the investment of \$1 in stocks A, B, and C will grow to $\$1(1+0.029)^5 = \$1.154$, $\$1(1+0.068)^5 = \$1.389$, and $\$1(1+0.030)^5 = \$1.159$, respectively. However, if an equally weighted (EW) portfolio is formed using these three stocks, the portfolio has a greater GMR than any of individual stocks. Specifically, the investment of \$1 in the portfolio will grow to $\$1(1+0.126)^5 = \$1.810$. The reason for this result is that the GMR is greater when returns are less volatile. The GMR is smaller when there are extreme returns, although the arithmetic mean return is larger.

The approximate relation between the geometric mean and the arithmetic mean is

$$\bar{r}_g \approx \bar{r} - \frac{1}{2}\sigma^2 \tag{10.6}$$

**TABLE 10.1** The GMR and Arithmetic Mean Return

| Period | Stocks | | | EW portfolio |
|---|---|---|---|---|
| | A | B | C | |
| 1 | 0.85 | −0.25 | −0.23 | 0.12 |
| 2 | −0.50 | 0.70 | 0.18 | 0.13 |
| 3 | 0.42 | −0.20 | 0.17 | 0.13 |
| 4 | −0.45 | 0.70 | 0.15 | 0.13 |
| 5 | 0.60 | −0.20 | −0.05 | 0.12 |
| Arithmetic avg. ($\bar{r}$) | 0.184 | 0.150 | 0.044 | 0.126 |
| Std. deviation ($\sigma$) | 0.621 | 0.502 | 0.180 | 0.006 |
| GMR ($\bar{r}_g$) | 0.029 | 0.068 | 0.030 | 0.126 |

where $\bar{r}$ is the arithmetic mean.[5] Thus, the geometric mean is always less than or equal to the arithmetic mean. It is seen from equation (10.6) that the greater the volatility ($\sigma^2$), the less the GMR. Thus, the GMR is usually higher for well-diversified portfolios.

## 10.2 THE SAFETY-FIRST CRITERION

The Markowitz portfolio theory usually follows the mean-variance approach in which investors are assumed to choose among portfolios on the basis of a utility function defined in terms of the mean and variance of the portfolio return. In the mean-variance approach, however, the utility function is arbitrarily selected. Because of the arbitrary nature of utility functions, there were attempts to depart from the utility framework and invoke criterion based on more objective goals. One of these attempts is the safety-first criteria.

*Safety-first* criteria for portfolio selection are concerned only with the risk of failing to achieve a certain minimum target return or secure prespecified safety margins. The risk is commonly expressed as

$$\text{Prob}\left(r_p \leq r_L\right) \leq \alpha \tag{10.7}$$

where $r_p$ is the return of portfolio $p$, $r_L$ is a certain desired level of return below which the investor does not wish to fall, which is often referred to as the disaster level or the safety threshold, and $\alpha$ is an acceptable limit on the probability of failing to earn the minimally acceptable level of return, $r_L$.

### 10.2.1 Roy's Safety-First Criterion

Roy (1952) developed a safety-first criterion that seeks to minimize the probability of earning a disaster level of return, $\alpha$ in equation (10.7).[6] That is,

$$\text{Minimize} \quad \text{Prob}\left(r_p < r_L\right) \tag{10.8}$$

Roy's safety-first criterion implies that investors choose their portfolios by minimizing the loss probability for a fixed safety threshold called the floor return. Roy's criterion tries to control risk for a fixed return whereas Markowitz's mean-variance criterion offers a menu of positively related pairs of points having both the maximum local return and minimum local risk.

Table 10.2 shows 50 rates of return from four portfolios through 50 common time periods. These returns are sorted in ascending order in the table. Suppose that the investor does not wish returns to fall below −1 percent. The frequencies of the returns that fall below $r_L = -1\%$ are 8, 11, 7, and 6 out of 50 return observations for portfolios A, B, C, and D, respectively. That is, the probabilities that the return of each portfolio falls below $r_L = -1\%$ are 16 percent (=8/50), 22 percent (=11/50), 14 percent (=7/50), and 12 percent (=6/50) for portfolios A, B, C, and D, respectively. With respect to Roy's criterion, therefore, portfolio D is the best because it has the smallest probability to earn returns lower than the prespecified minimally acceptable level of return. Portfolios C, A, and B are the next best in order. That is, D > C > A > B. This preference ranking can differ from rankings based on a mean-variance approach such as the Sharpe ratio (to be explained further in Chapter 18). The Sharpe ratio, denoted $S_p$, is defined as the ratio of the excess return to the standard deviation.

$$S_p = \frac{\bar{r}_p - r_f}{\sigma_p} \tag{10.9}$$

Assuming the risk-free rate of return is $r_f = 0.012\%$, the Sharpe ratios of these four portfolios over 50 periods are 0.225, −0.110, 0.145, and 0.088, respectively. Thus, the ranking with respect to the Sharpe ratio is. $A > C > D > B$.

If the statistical distribution of returns can be characterized by mean and variance, Roy's safety-first criterion can be analyzed in a mean-variance context. For analytical convenience, we assume that returns are normally distributed. Then, the optimum portfolio is the one that has the smallest area of a left-hand side tail of the normal distribution. The probability in equation (10.8) can be transformed as

$$\begin{aligned}\text{Prob}\left(r_p < r_L\right) &= \text{Prob}\left[\left(\frac{r_p - \text{E}\left(r_p\right)}{\sigma_p}\right) < \left(\frac{r_L - \text{E}\left(r_p\right)}{\sigma_p}\right)\right] \\ &= \text{Prob}\left[Z < \left(\frac{r_L - \text{E}\left(r_p\right)}{\sigma_p}\right)\right] \end{aligned} \tag{10.10}$$

where Z is a standard normal variate. Thus,

$$\text{Minimize Prob}\left(r_p < r_L\right) \equiv \text{Minimize Prob}\left[Z < \left(\frac{r_L - \text{E}\left(r_p\right)}{\sigma_p}\right)\right] \tag{10.11}$$

In Figure 10.1, the shaded area under the standard normal distribution indicates the probability that the return of a portfolio falls below the safety threshold level, $r_L$. The probability that a standard normal variate is less than $\left[r_L - \text{E}\left(r_p\right)\right]/\sigma_p$ can be easily calculated from the standard normal distribution table.

**TABLE 10.2** Daily Returns from Four Stocks

| Day | Unsorted original returns (%) A | B | C | D | Sorted returns in ascending order (%) A | B | C | D |
|---|---|---|---|---|---|---|---|---|
| 1 | 0.80 | 0.40 | −1.59 | −0.34 | −1.99 | −4.31 | −1.90 | −2.59 |
| 2 | 0.04 | 1.54 | 1.51 | 1.11 | −1.91 | −3.10 | −1.84 | −1.49 |
| 3 | 5.38 | −1.17 | −1.90 | −0.84 | −1.74 | −2.09 | −1.59 | −1.41 |
| 4 | 2.36 | −0.46 | −0.21 | −1.28 | −1.66 | −1.43 | −1.26 | −1.28 |
| 5 | −0.31 | 2.29 | 0.45 | 0.49 | −1.55 | −1.39 | −1.24 | −1.21 |
| 6 | −1.99 | −1.43 | 0.70 | 0.09 | −1.37 | −1.17 | −1.11 | −0.98 |
| 7 | 0.71 | 0.16 | 1.13 | 4.04 | −1.26 | −1.11 | −1.02 | −0.96 |
| 8 | −1.74 | −3.10 | −1.84 | −2.59 | −1.10 | −1.07 | −0.73 | −0.92 |
| 9 | 0.54 | 0.67 | −0.04 | 1.64 | −0.77 | −1.06 | −0.57 | −0.84 |
| 10 | −1.26 | −0.57 | 0.50 | −0.63 | −0.74 | −1.04 | −0.51 | −0.63 |
| 11 | 1.93 | −0.61 | 0.11 | 0.77 | −0.63 | −1.03 | −0.48 | −0.62 |
| 12 | 1.46 | 1.68 | 1.49 | 2.44 | −0.54 | −0.82 | −0.45 | −0.56 |
| 13 | 0.18 | 0.11 | 0.76 | 0.94 | −0.40 | −0.75 | −0.37 | −0.42 |
| 14 | 1.65 | 0.95 | 2.03 | 1.33 | −0.31 | −0.65 | −0.35 | −0.41 |
| 15 | 0.07 | −0.33 | 0.72 | 0.02 | −0.30 | −0.61 | −0.24 | −0.34 |
| 16 | 0.38 | 0.41 | 0.22 | 0.60 | −0.30 | −0.57 | −0.21 | −0.34 |
| 17 | 0.82 | −1.39 | −0.73 | −1.49 | −0.10 | −0.54 | −0.21 | −0.21 |
| 18 | 0.92 | 0.79 | 0.61 | 0.51 | −0.07 | −0.46 | −0.19 | −0.21 |
| 19 | −0.30 | 2.70 | 0.93 | 1.07 | 0.04 | −0.45 | −0.17 | −0.06 |
| 20 | 2.00 | 0.81 | 0.33 | 0.06 | 0.07 | −0.36 | −0.04 | −0.05 |
| 21 | 0.37 | 0.32 | −0.37 | 0.16 | 0.17 | −0.35 | 0.05 | −0.05 |
| 22 | −1.10 | −0.82 | −0.48 | −0.42 | 0.18 | −0.33 | 0.11 | 0.00 |
| 23 | −0.54 | −0.36 | −0.45 | −0.56 | 0.26 | −0.33 | 0.17 | 0.00 |
| 24 | 1.08 | 1.77 | 0.98 | 1.50 | 0.27 | −0.31 | 0.19 | 0.02 |
| 25 | −0.10 | 0.36 | −0.21 | 0.11 | 0.32 | −0.22 | 0.22 | 0.02 |
| 26 | −0.40 | 0.66 | −0.51 | 0.11 | 0.37 | −0.10 | 0.26 | 0.06 |
| 27 | −1.91 | −2.09 | −1.24 | −0.62 | 0.38 | −0.01 | 0.27 | 0.06 |
| 28 | 0.65 | 0.27 | 0.52 | −0.21 | 0.47 | 0.09 | 0.33 | 0.06 |
| 29 | 2.04 | 1.29 | 1.26 | 0.90 | 0.50 | 0.11 | 0.40 | 0.09 |
| 30 | −0.77 | −0.33 | −0.57 | 0.45 | 0.53 | 0.16 | 0.44 | 0.11 |
| 31 | 0.17 | −1.06 | 0.27 | −0.98 | 0.54 | 0.27 | 0.45 | 0.11 |
| 32 | 0.50 | −1.04 | −0.24 | 0.06 | 0.54 | 0.32 | 0.50 | 0.16 |
| 33 | −0.63 | −0.65 | −0.17 | −0.21 | 0.55 | 0.36 | 0.52 | 0.16 |
| 34 | −0.74 | −1.11 | −0.19 | −0.96 | 0.65 | 0.40 | 0.55 | 0.45 |
| 35 | 0.47 | −0.22 | 1.25 | 0.60 | 0.71 | 0.41 | 0.58 | 0.49 |
| 36 | 0.54 | −0.54 | 0.74 | 0.00 | 0.71 | 0.44 | 0.61 | 0.51 |
| 37 | −0.07 | 0.59 | 0.26 | 0.16 | 0.80 | 0.59 | 0.70 | 0.54 |
| 38 | 0.87 | −4.31 | 0.19 | 1.07 | 0.82 | 0.59 | 0.72 | 0.60 |
| 39 | −0.30 | −0.75 | −1.11 | −1.41 | 0.87 | 0.61 | 0.74 | 0.60 |
| 40 | 0.27 | −1.07 | 0.17 | 0.06 | 0.92 | 0.66 | 0.76 | 0.77 |
| 41 | −1.66 | −0.31 | −1.02 | −0.92 | 0.98 | 0.67 | 0.93 | 0.90 |
| 42 | 2.57 | −0.01 | 0.40 | −0.06 | 1.08 | 0.79 | 0.98 | 0.94 |
| 43 | 0.53 | 0.44 | 0.58 | −0.41 | 1.46 | 0.81 | 0.99 | 1.07 |
| 44 | 0.98 | 0.09 | 0.99 | 0.00 | 1.65 | 0.95 | 1.13 | 1.07 |
| 45 | 0.32 | −0.45 | 0.05 | 0.02 | 1.93 | 1.29 | 1.25 | 1.11 |
| 46 | 0.26 | 0.59 | 0.44 | −0.05 | 2.00 | 1.54 | 1.26 | 1.33 |
| 47 | 0.55 | 0.61 | 1.33 | −0.05 | 2.04 | 1.68 | 1.33 | 1.50 |
| 48 | 0.71 | −0.35 | −0.35 | 0.54 | 2.36 | 1.77 | 1.49 | 1.64 |
| 49 | −1.37 | −0.10 | 0.55 | −0.34 | 2.57 | 2.29 | 1.51 | 2.44 |
| 50 | −1.55 | −1.03 | −1.26 | −1.21 | 5.38 | 2.70 | 2.03 | 4.04 |
| Avg | 0.31 | −0.12 | 0.14 | 0.11 | | | | |
| Std dev | 1.31 | 1.23 | 0.88 | 1.06 | | | | |

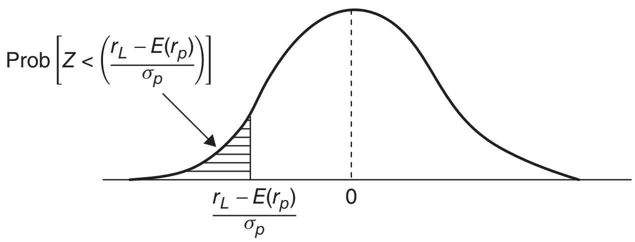


**FIGURE 10.1** Probability the Return Falls Below $r_L$

## EXAMPLE 10.1

The expected returns and standard deviations of the four portfolios from Table 10.2 are shown in Table 10.3. Assuming the returns of these portfolios are normally distributed and the threshold safety level or the floor return ($r_L$) is 3 percent, which portfolio is best with respect to Roy's safety-first criterion?

**TABLE 10.3** Expected Return and Standard Deviation of Four Portfolios

| | Portfolios | | | |
|---|---|---|---|---|
| | A | B | C | D |
| Expected return, $E(r_p)$ | 12% | 10% | 15% | 10% |
| Standard deviation, $\sigma_p$ | 9% | 4% | 10% | 8% |
| $\left[r_L - E(r_p)\right]/\sigma_p$ | −1.00 | −1.75 | −1.20 | −0.875 |
| Prob $\left[Z < \left(\frac{r_L - E(r_p)}{\sigma_p}\right)\right]$ | 0.1587 | 0.0401 | 0.1151 | 0.1908 |

The probability that the return of portfolio A is less than $r_L = 3\%$ is Prob $(Z < -1.00)$. From the standard normal distribution table, this probability is 0.1587. Similarly, the probabilities for portfolios B, C, and D are 0.0401, 0.1151, and 0.1908, respectively. The return from portfolio B has the smallest probability of falling below 3 percent and, thus, is the best with respect to Roy's criterion.

Roy's safety-first criterion is related to the Sharpe ratio. Minimizing the probability in equation (10.11) is equivalent to

$$\text{Minimize Prob}\left[Z < \left(\frac{r_L - E(r_p)}{\sigma_p}\right)\right] \equiv \text{Minimize}\left[\frac{r_L - E(r_p)}{\sigma_p}\right]$$

$$\equiv \text{Maximize}\left[\frac{E(r_p) - r_L}{\sigma_p}\right] \qquad (10.12)$$

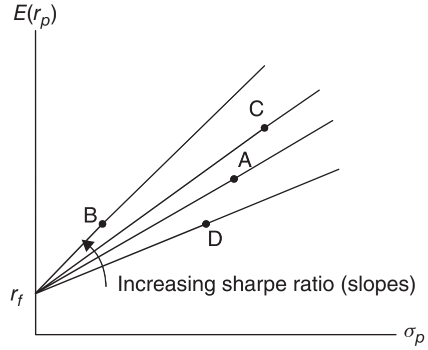


**FIGURE 10.2** Sharpe's Criterion Applied to Table 10.3

Thus, the optimum portfolio is the one with the greatest ratio $\left[E\left(r_p\right) - r_L\right]/\sigma_p$. Note that this decision criterion can be applied to any statistical distribution, as long as this distribution is characterized by only two parameters: mean and variance. The normality assumption of asset returns is not necessary. If the safety threshold $r_L$ is replaced by the risk-free return $r_f$ and we let the ratio be $S_p$, then

$$\text{Sharpe ratio} = S_p = \frac{E\left(r_p\right) - r_f}{\sigma_p} \tag{10.9}$$

Rewriting equation (10.9) yields

$$E\left(r_p\right) = r_f + S_p\sigma_p. \tag{10.13}$$

This is a straight line equation in $\left[\sigma_p, E\left(r_p\right)\right]$ space with an intercept $r_f$ and slope $S_p$. Thus, the optimum portfolio is the one that has the greatest slope of the straight line originating $r_f$ and passing through the portfolio. Figure 10.2 illustrates four lines representing portfolios A, B, C, and D in Table 10.3 of Example 10.1, respectively. The slope of the line indicates the desirability of the portfolio. The greater the slope, the more the desirability and the less probability that the return falls below the floor return. Portfolios lying on the same line are equally desirable, because they have the same slope. By combining the risk-free asset (with borrowing or lending at the risk-free rate) with a risky portfolio lying on the line, many equally desirable portfolios can be created. The line passing through $r_f$ and point B has the highest slope; thus, portfolio B is most derirable. The remaining portfolio rankings are C > A > D.

### 10.2.2 Kataoka's Safety-First Criterion

Kataoka (1963) developed a safety-first criterion in which investors choose the portfolio with an insured return, $r_L$, as high as possible subject to the following constraint:[7] the probability that the portfolio return is no greater than the insured return must not exceed a predetermined level, denoted $\alpha$.[8] In other words, the problem statement of Kataoka's safety-first criterion is

$$\begin{aligned} &\text{Maximize } r_L \\ &\text{subject to Prob}\left(r_p < r_L\right) \le \alpha \end{aligned} \tag{10.14}$$

## EXAMPLE 10.2

Which portfolio is the most desirable among the four portfolios listed in Table 10.3 of Example 10.1 with respect to Kataoka's criterion, when $\alpha = 10\%$?

In Kataoka's criterion, investors choose the portfolio with the greatest insured return, $r_L$, under the constraint $\text{Prob}\left(r_p < r_L\right) \le 0.10$. By looking at the sorted returns in ascending order, we can observe that the insured return for portfolio A is −1.37 percent, because the frequency of portfolio A's returns being not greater than −1.37 percent is less than or equal to 5 out of 50 observations (or equivalently, 10 percent). That is, the constraint $\text{Prob}\left(r_A < -1.37\right) \le 0.10$ is satisfied. In this manner, we can find the insured levels of return satisfying the constraint for portfolios B, C, and D are −1.17 percent, −1.11 percent, and −0.98 percent, respectively. The maximum value among these insured levels of return is $L_D = -0.95$ percent. Thus, portfolio D is the best with respect to Kataoka's safety-first criterion, and portfolios C, B, and A are the next most desirable.

Like Roy's criterion, Kataoka's safety-first criterion can also be analyzed in a mean-variance approach, if the statistical distribution of returns can be characterized in terms of the mean and variance. The insured level of return, $r_L$, is the abscissa value below which the area under the statistical distribution is less than or equal to the predetermined level $\alpha$. Figure 10.3 shows the insured level of return, $r_L$, when returns are normally distributed.

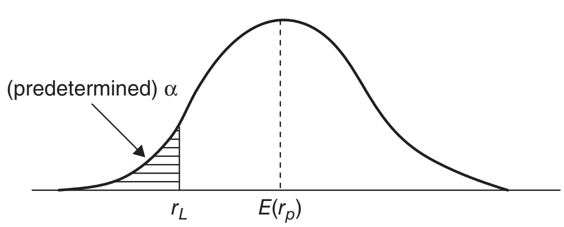


**FIGURE 10.3** The Insured Level $\left(r_L\right)$ Satisfying the Constraint $\text{Prob}(r_p < r_L) \le \alpha)$

## EXAMPLE 10.3

Table 10.3 presents the expected returns and standard deviations of four portfolios. Assume the portfolios' returns are normally distributed and the predetermined level of $\alpha$ is 5 percent. Which portfolio is the most desirable with respect to Kataoka's criterion?

The constraint can be transformed to

$$\text{Prob}\left(r_p < r_L\right) \leq \alpha \quad \Longleftrightarrow \quad \text{Prob}\left[Z \leq \left(\frac{r_L - E\left(r_p\right)}{\sigma_p}\right)\right] \leq \alpha \tag{10.15}$$

Since $\alpha = 5\%$, the abscissa value in the standard normal distribution is $\left[r_L - E\left(r_p\right)\right]/\sigma_p = -1.645$. Thus, the insured level of return equals

$$r_L = E\left(r_p\right) + (-1.645) \times \sigma_p \tag{10.16}$$

The insured level of return for portfolio A is

$$r_{L_A} = 12 + (-1.645) \times 9 = -2.805\%.$$

That is, the probability that portfolio A's return is less than −2.805 is at most 5 percent. Likewise, the insured levels of return for portfolios B, C, and D are

$$r_{L_B} = 10 + (-1.645) \times 4 = 3.42\%,$$

$$r_{L_C} = 15 + (-1.645) \times 10 = -1.45\%,$$

$$r_{L_D} = 10 + (-1.645) \times 8 = -3.16\%$$

At the same loss probability of $\alpha = 5\%$, the lowest possible acceptable returns for portfolios A, B, C, and D are −2.805 percent, 3.42 percent, −1.45 percent, and −3.16 percent, respectively. Thus, portfolio B is most desirable with respect to Kataoka's criterion, and portfolios C, A, and D are the next most desirable, in order of desirability.

Kataoka's safety-first criterion is also related to the traditional mean-variance approach in one circumstance. If the returns are normally distributed, then the constraint of equation (10.14) can be transformed as

$$\text{Prob}\left(r_p < r_L\right) \leq \alpha \quad \Longleftrightarrow \quad \left(\frac{r_L - E\left(r_p\right)}{\sigma_p}\right) \leq -z_\alpha \quad \Longleftrightarrow \quad r_L \leq E\left(r_p\right) - z_\alpha \sigma_p \tag{10.17}$$

where $z_\alpha$ is chosen so that $\text{Prob}\left(Z > z_\alpha\right) = \alpha$. Because investors want to make the insured level of return as large as possible, the last constraint in equation (10.17) becomes

$$r_L = E\left(r_p\right) - z_\alpha \sigma_p \tag{10.18}$$

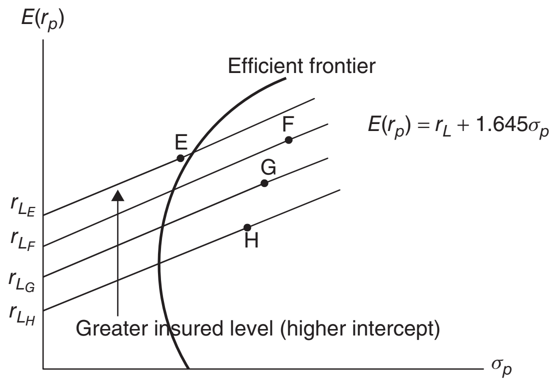


**FIGURE 10.4** Karaoka's Safety-first Criterion

If $\alpha = 5\%$, then $z_\alpha = 1.645$, and equation (10.18) becomes

$$E\left(r_p\right) = r_L + 1.645\sigma_p \tag{10.19}$$

Figure 10.4 illustrates the parallel straight lines of equation (10.19) passing through portfolios E, F, G, and H in $\left[\sigma_{\mathrm{p}}, E(r_p)\right]$ space, and they are in the investment opportunity set. These lines have the same slope of 1.645. Because the intercept indicates the insured level of return for each portfolio, portfolios locating on the straight line with the greater intercept are more desirable with respect to Kataoka's safety-first criterion. Thus, portfolio E is the best, because the straight line passing through the location of portfolio E has the highest intercept. Portfolios F, G, and H are the next best, respectively.

When Kataoka's safety-first criterion is contrasted with Roy's safety-first criterion in $\left[\sigma_{\mathrm{p}}, E(r_p)\right]$ space, we see that Roy's criterion favors portfolios with a greater slope from the fixed intercept, whereas Kataoka's criterion prefers portfolios with the greatest intercept for a given slope.

### 10.2.3 Telser's Safety-First Criterion

Telser (1955) introduced another safety-first criterion three years after Roy's safety-first criterion.[9] His criterion assumes that investors maximize expected return, subject to the constraint that the probability of a return less than or equal to a prespecified minimum disaster level ($r_L$) does not exceed a given probability. Mathematically, Telser's criterion is expressed as

$$\begin{aligned} &\text{Maximize} \quad E(r_p) \\ &\text{subject to Prob}\left(r_p \le r_L\right) \le \alpha \end{aligned} \tag{10.20}$$

In the constraint of equation (10.20), the minimum disaster level, $r_L$, and the loss probability, $\alpha$, are prespecified. Like equation (10.17), the constraint used in Telser's criterion of equation (10.20) can be reformulated as

$$E(r_p) \ge r_L + 1.645\sigma_p \tag{10.21}$$

Among the portfolios satisfying this constraint, the optimum portfolio with respect to Telser's criterion is the one with the greatest expected return.

## EXAMPLE 10.4

Table 10.3 presents the expected returns and standard deviations from four portfolios. For this example, assume the portfolio's returns are normally distributed. If the minimum disaster level of return is $r_L = -2\%$ and the loss probability is $\alpha = 5\%$, which portfolio is the most desirable under Telser's criterion?

The constraints for the four portfolios are

$$E\left(r_A\right) \geq -2 + 1.645 \times 9 = 12.81\%,$$
$$E\left(r_B\right) \geq -2 + 1.645 \times 4 = 4.58\%,$$
$$E\left(r_C\right) \geq -2 + 1.645 \times 10 = 14.45\%,$$
$$E\left(r_D\right) \geq -2 + 1.645 \times 8 = 11.16\%.$$

The expected returns of portfolios A and D do not satisfy the constraint. The expected returns of portfolios B and C satisfy the constraint. We choose the portfolio that has the greatest expected return. Thus, portfolio C is most desirable with respect to Telser's safety-first criterion. Portfolio C is the second most desirable. Table 10.4 summarizes the statistics needed for Roy's, Kataoka's, and Telser's criteria.

**TABLE 10.4** Three Safety-first Criteria for the Hypothetical Portfolios

| | Portfolios | | | |
|---|---|---|---|---|
| | A | B | C | D |
| Expected return, $E(r_p)$ | 12% | 10% | 15% | 10% |
| Standard deviation, $\sigma_p$ | 9% | 4% | 10% | 8% |
| Roy's: Prob $\left\{Z \leq \left[\left(r_L - E(r_p)\right)/\sigma_p\right]\right\}$ | 0.1587 | 0.0401 | 0.1151 | 0.1908 |
| Kataoka's: Insured return $(r_L)$ with $\alpha = 5\%$ | −2.85% | 3.40% | 1.90% | −1.30% |
| Telser's: $r_L + 1.645\sigma_p$ if $L = -2\%$ | 12.81% | 4.58% | 14.45% | 11.16% |

Like the two previous safety-first criteria, Telser's criterion can also be related to the traditional mean-variance approach under one circumstance. If the returns are normally distributed and there exists no risk-free asset, then the constraint in Telser's criterion can be illustrated as in Figure 10.5. Portfolios satisfying the constraint must lie in the shaded area, which is the intersection between the constraint and the opportunity set inside the mean-variance efficient frontier. Thus, the optimum portfolio is the one that has the greatest expected return among the portfolios located in the shaded region. In fact, the optimum portfolio will be on the mean-variance efficient frontier.

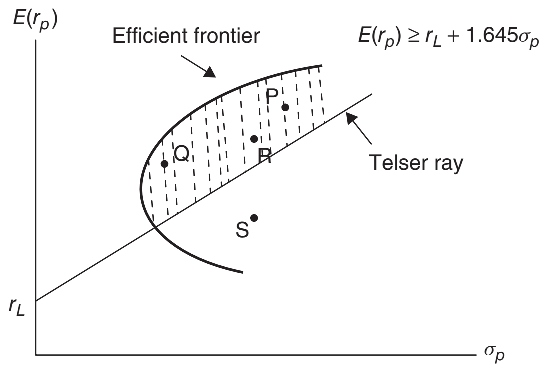


**FIGURE 10.5** Telser's Safety-first Criterion without the Risk-free Asset

If a risk-free asset exists, the efficient frontier will equal the capital market line (CML). The intersection of the set satisfying the constraint and the opportunity set is the shaded area in Figure 10.6. Among the portfolios located in the shaded area, portfolio Q has the greatest expected return, at the intersection point of the Telser ray and the CML. Thus, portfolio Q is best with respect to Telser's criterion. The expected return of portfolio Q satisfies

$$r_L = E\left(r_Q\right) - z_\alpha \sigma_Q \tag{10.22}$$

Meanwhile, portfolio Q is on the CML and is formed by investing $w_m$ in the market portfolio, $m$, and $\left(1 - w_m\right)$ in the risk-free asset. Thus, equation (10.22) can be rewritten as

$$r_L = \left[w_m E\left(r_m\right) + \left(1 - w_m\right) r_f\right] - z_\alpha \left(w_m \sigma_m\right) \tag{10.23}$$

Then,

$$w_m = \frac{r_L - r_f}{E\left(r_m\right) - r_f - z_\alpha \sigma_m} \tag{10.24}$$

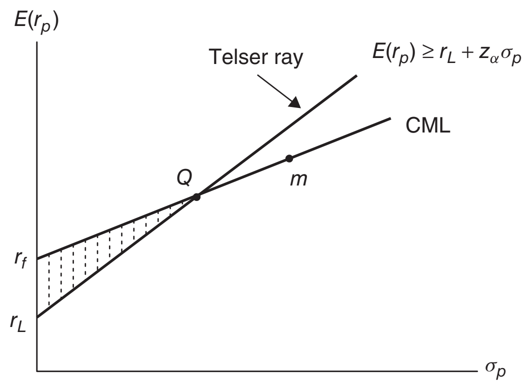


**FIGURE 10.6** Telser's Safety-first Criterion with the Risk-free Asset

If the investor sets the loss probability ($\alpha$) to be very small (i.e., very conservative), then $z_\alpha$ is large, and so the Telser ray will have a large positive slope and intersect the CML close to the point of the risk-free return. If the minimum disaster level of return ($r_L$) is close to the risk-free return, then the same result holds.

In summary, Kataoka's safety-first criterion contrasts with Roy's safety-first criterion in that the optimum portfolio with respect to Roy's criterion is the one with the smallest loss probability at a prespecified level of the insured return, $r_L$. And the optimum portfolio with respect to Kataoka's criterion is the one with the largest insured level of return at a prespecified level of loss probability, $\alpha$. In Telser's criterion, both the insured return or the minimum disaster level of return and the loss probability are prespecified. Then, among the portfolios satisfying the prespecified $r_L$ and $\alpha$, the optimum portfolio is the one with the greatest expected return.

## 10.3 STOCHASTIC DOMINANCE CRITERIOS

Portfolio analysis with stochastic dominance can be formulated differently to accommodate different utility functions.

### 10.3.1 First- Order Stochastic Dominance

A selection criterion called the *first -order stochastic dominance criterio* is a simple form of analysis that competes with Markowitz´s mean-variance portfolio.

analysis. Stochastic dominance focuses on every bit of information in the probability distributions rather than simply focusing on the probability distributions' first two moments. As a result, stochastic dominance selection rules will occasionally yield portfolios that are more desirable than Markowitz-efficient portfolios.

When empirical distributions are non-normal, there exists the potential for serious conflicts between choices made by the stochastic dominance criterion and those made by the mean-variance criterion. Figure 10.12 shows the probability distributions for three risky investments. Uniform probability distributions are used to expedite the explanation, but stochastic dominance rules apply to any probability distribution. The expected returns on the three assets are $\mu_A = 5\%$, $\mu_B = 20\%$, and $\mu_C = 20\%$. Their standard deviations are $\sigma_A = 8.66\%$, $\sigma_B = 11.55\%$, and $\sigma_C = 14.43\%$, respectively.[10] Figure 10.13 graphically depicts these three assets on $(\sigma, \mu)$ space.

In the mean-variance portfolio analysis, assets A and B are efficient, while asset C is inefficient. However, inefficient asset C is more desirable than efficient asset A in some circumstances. The probability that asset A earns less than −5 percent is 1/6, while this probability for asset C is zero.[11] Or, the probability that asset A earns less than zero is 1/3, while this probability for asset C is only 1/10. For any given specific rate of return that is minimally tolerable, the chance of earning less than the specific rate of return is greater for asset A than for asset C. To investors who prefer more to less, asset C is more desirable than asset A, even though asset C is inefficient. This demonstrates that portfolio analysis that considers only the first two moments (mean and variance) may waste some information that could be used to maximize

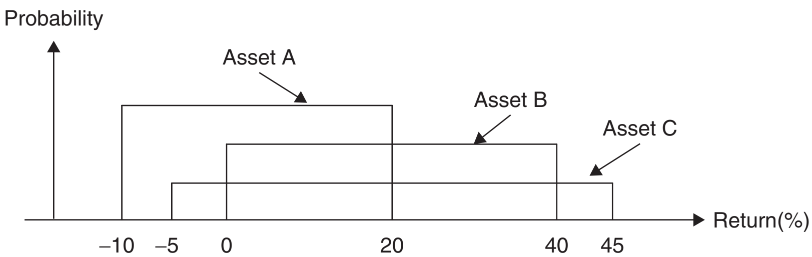


**FIGURE 10.12** Uniform Probability Distributions for Three Assets' Returns

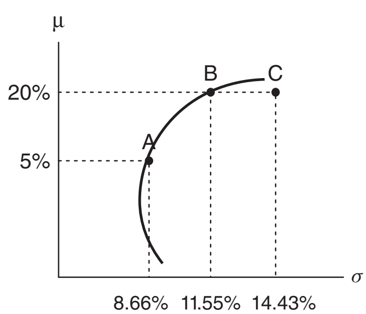


**FIGURE 10.13** Three Assets in $(\sigma, \mu)$ Space

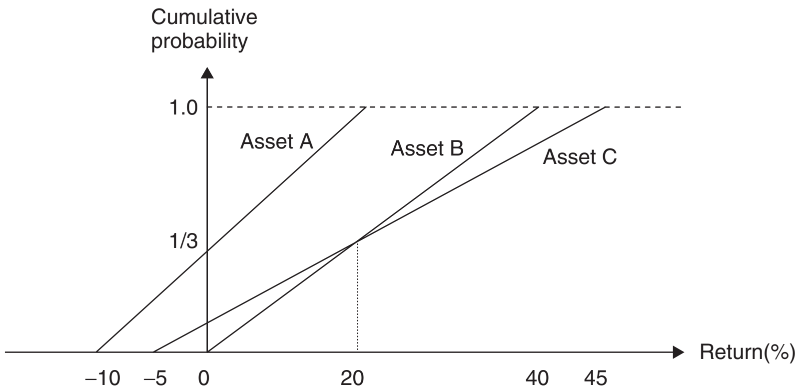


**FIGURE 10.14** Cumulative Probability Distributions of Returns for Three Assets

investors' expected utility. Figure 10.14 illustrates the cumulative density function (cdf) or the cumulative distribution of the return for each of the three assets. Over all possible values of the returns, the cumulative probability that the return from asset A is less than a specific return is greater than the cumulative probability that the return from asset C is less than the specific return. In other words, the probability of earning less than any given amount of return from asset A is less than for asset C. In consideration of the downside risk, therefore, asset C is more desirable than asset A. Likewise, asset B is also more desirable than asset A.

In general, the cumulative probability distribution, $F(x)$, is defined as

$$F(r) = \text{Prob}(X \leq r) = \int_{-\infty}^{r} f(x)\,dx \tag{10.46}$$

where $f(r)$ is the probability density function (pdf) of asset returns. The cumulative distribution indicates the probability that an asset earns less than a certain level of return, $x$. Figure 10.15 illustrates the cumulative probability that the return is less than or equal to a given level of return (the shaded area) for that particular statistical distribution.

Figure 10.16 illustrates an example of the first-order stochastic dominance (FSD). The cumulative distribution for asset A lies below that for asset B for any given value of the return. It is said that asset A is more desirable than asset B in FSD.

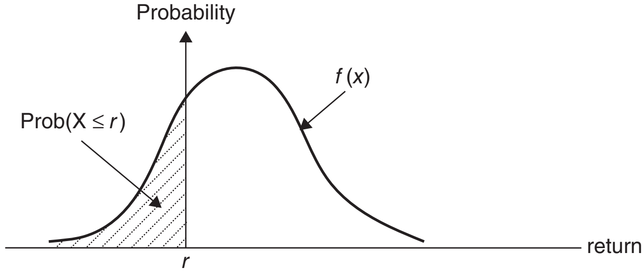


**FIGURE 10.15** Cumulative Probability

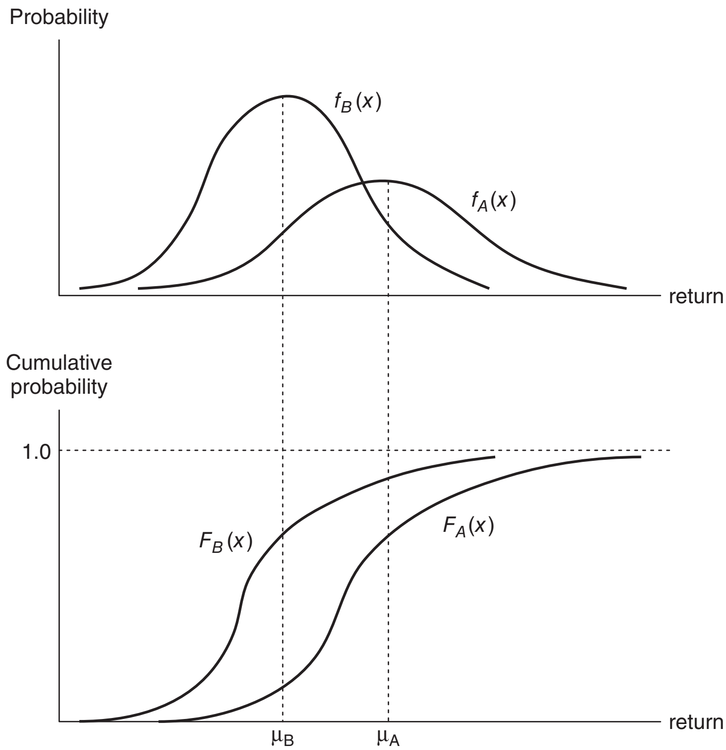


**FIGURE 10.16** Illustrations of First-order Stochastic Dominance

***FSD rule:*** Under the condition that investors prefer more to less (equivalently, $U'(r) > 0$), asset A dominates asset B in FSD if and only if $F_A(r) \leq F_B(r)$ or equivalently, $F_A(r) - F_B(r) \leq 0$ for all $r$ with the strict inequality holding for at least one value of $r \in [a, b]$, where $F_A(\cdot)$ and $F_B(\cdot)$ are the cumulative distribution of the return of assets A and B, respectively, and $a$ and $b$ are the lower and upper bounds for $r$.

The economic interpretation of the FSD rule is that for all monotonic *nondecreasing* utility functions (i.e., $U'(r) > 0$), the expected utility from asset A is no less than that from asset B for *all* investors, irrespective of their attitudes toward risk (i.e., $\mathrm{E}[U_A(r)] \geq \mathrm{E}[U_B(r)]$). Furthermore, regardless of the assumptions about the utility functions, $F_A(r) \leq F_B(r)$ implies that the expected return of asset A is greater than or equal to the expected return from asset B [i.e., $E_A(r) \geq E_B(r)$ for positive $r$]. Note that the positive value restriction does not affect this implication because $r$ can be simply replaced with $1 + r$. The proofs for these assertions are in the appendix to this chapter.

In fact, the following three statements are equivalent:

1. Asset A dominates asset B in FSD.
2. $F_A(r) \leq F_B(r)$, where $r \in [a, b]$.
3. $r_A =^{\text{dist}} r_B + \tilde{\upsilon}$, where a random variable $\tilde{\upsilon} \geq 0$. "$=^{\text{dist}}$" means that $r_A$ and $r_B + \tilde{\upsilon}$ are equal *in distribution*.

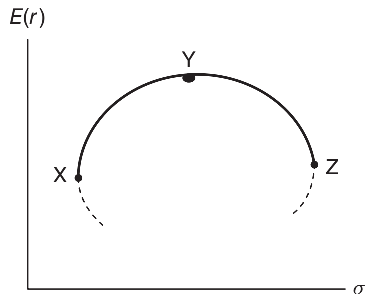


**FIGURE 10.17** Efficient Set by First-order Stochastic Dominance

The third statement means that if asset A dominates asset B in FSD, the expected return of asset A is at least as high as the expected return from asset B. However, the converse is not true.

First-order stochastic dominance can be related to the mean-variance analysis if the returns are normally distributed. First-order stochastic dominance implies that investors prefer a higher expected return at any level of standard deviation. Thus, portfolios produced by first-order stochastic dominance lie on the upper part of the investment opportunity set. Figure 10.17 illustrates the efficient set by first-order stochastic dominance. Portfolios on the curve XYZ in Figure 10.16 have greater expected return than any other portfolio below the curve XYZ; thus, they are FSD-efficient. However, portfolios on the segment YZ are not mean-variance efficient, even though they are FSD-efficient.

## EXAMPLE 10.5

Assets A and B have the outcomes with the associated probabilities shown in Table 10.5. Does asset A dominate asset B in FSD?

**TABLE 10.5** Possible Outcomes of Two Assets

| Asset A | | Asset B | |
|---|---|---|---|
| **Probability** | **Return** | **Probability** | **Return** |
| 1/4 | 4 | 1/4 | 3 |
| 1/4 | 5 | 1/4 | 4 |
| 1/4 | 7 | 1/4 | 6 |
| 1/4 | 8 | 1/4 | 7 |

The cumulative probabilities of the two assets are shown in Table 10.6.

**EXAMPLE 10.5** *(Continued)*

**TABLE 10.6** Cumulative Probabilities for Two Assets

| Returns ($x$) | Cumulative probability Prob ($r \le x$) A | B |
|---|---|---|
| 3 | 0 | 1/4 |
| 4 | 1/4 | 2/4 |
| 5 | 2/4 | 2/4 |
| 6 | 2/4 | 3/4 |
| 7 | 3/4 | 1 |
| 8 | 1 | 1 |

Table 10.6 shows that the cumulative probability of the return from asset A is no greater than that from asset B for any value of the return. This fact is graphically shown in Figure 10.18. The cumulative distribution for asset B (solid line) never lies below that for asset A (dotted line). Thus, asset A dominates asset B in FSD.

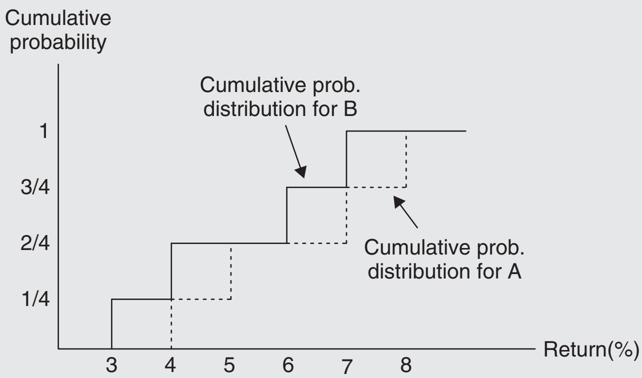


**FIGURE 10.18** Cumulative Probability Distributions for Two Assets

## 10.3.2 Second-Order Stochastic Dominance

In the previous section's Figures 10.13 and 10.14, any preference between assets B and C cannot be made in first-order stochastic dominance because their cumulative distributions cross (overlap). Specifically, the probability of earning less than the return between −5 percent and 20 percent from asset B is lower than from asset C, whereas the probability of earning less than the return between 20 percent and 45 percent from asset C is lower than from asset B. In other words, asset B is more desirable than asset C over the low-return region, whereas asset C is more desirable than asset B over the high-return region. If the assumption of risk aversion is added to the FSD decision criterion, then a preference can be made between assets B and C. Risk-averse investors would be happier with a one percent increase from low returns than a one percent increase from high returns. Thus, asset B is more desirable than asset C to risk-averse investors. In this case, it is said that that asset B dominates asset C in second-order stochastic dominance (SSD).

***SSD rule:*** Under the conditions that

1. investors prefer more to less (equivalently, $U'(r) > 0$),
2. investors are risk averse (equivalently, $U''(r) < 0$), and
3. the expected returns of assets A and B are equal [i.e., $E_A(r) = E_B(r)$], asset A dominates asset B in SSD if and only if

$$\int_a^r F_A(x)\,dx \le \int_a^r F_B(x)\,dx \text{ or equivalently, } \int_a^r [F_A(x) - F_B(x)]\,dx \le 0$$

for all $r$ with the strict inequality holding for at least one value of $r \in [a, b]$, where $a$ and $b$ are the lower and upper bounds for $r$.

SSD implies that the expected utility from asset A exceeds or is equal to that from asset B (i.e., $\mathrm{E}[U_A(r)] \ge \mathrm{E}[U_B(r)]$) for all utility functions satisfying the conditions $U'(r) > 0$ and $U''(r) < 0$. See the end-of-chapter appendix for proof. The SSD rule can be applied when the cumulative probability distributions overlap as in Figure 10.14. However, the cumulative difference between the two cumulative probability distributions must remain nonnegative over the entire range of returns.

The relationship between the FSD and SSD efficient sets is that the SSD efficient set is a subset of the FSD efficient set. In other words, all the investments included in the SSD efficient set are also included in the FSD efficient set. However, the reverse does not hold. If asset A dominates asset B in SSD, then asset A also dominates asset B in FSD.

As with SSD, the following three statements are equivalent:

1. Asset A dominates asset B in SSD.
2. $E_A(r) = E_B(r)$ and $\int_a^b F_A(x)\,dx \le \int_a^b F_B(x)\,dx$ , where $r \in [a, b]$.
3. $r_B \stackrel{\text{dist}}{=} r_A + \tilde{\varepsilon}$, with $E(\tilde{\varepsilon}|r_A) = 0$.

The third statement means that $\mathrm{Var}(r_B) \ge \mathrm{Var}(r_A)$, because $E(\tilde{\varepsilon}|r_A) = 0$ implies $\mathrm{Cov}(r_A, \tilde{\varepsilon}) = 0$. Thus, if asset A dominates asset B in SSD, $E_A(r) = E_B(r)$ and $\mathrm{Var}(r_B) \ge \mathrm{Var}(r_A)$. However, the converse does not hold. Therefore, if one of the preceding three statements is satisfied, asset B is said to be more risky than asset A.[13]

Second-order stochastic dominance can be related to mean-variance analysis if returns are normally distributed. Portfolios on the segment XY in Figure 10.17 are SSD efficient because they satisfy the previous statements. Thus, the SSD-efficient set is equal to the mean-variance-efficient set if returns are normally distributed.

## EXAMPLE 10.6

Assets A and B have the probabilistic outcomes shown in Table 10.7. Does asset A dominate asset B in SSD?

**TABLE 10.7** Possible Outcomes of Two Assets

| Asset A | | Asset B | |
|---|---|---|---|
| Probability | Return | Probability | Return |
| 1/4 | 4 | 1/4 | 3 |
| 1/2 | 5 | 1/4 | 4 |
| 1/4 | 8 | 1/4 | 6 |
| | | 1/4 | 7 |

The cumulative probabilities and the accumulated values of the cumulative probabilities for the two assets are shown in Table 10.8.

**TABLE 10.8** Cumulative Probabilities for Assets A and B

| Returns ($x$) | Cumulative probabilities | | Accumulations of cumulative probabilities | |
|---|---|---|---|---|
| | $F_A(x)$ | $F_B(x)$ | Cum $[F_A(x)]$ | Cum $[F_B(x)]$ |
| 3 | 0 | 1/4 | 0 | 1/4 |
| 4 | 1/4 | 2/4 | 1/4 | 3/4 |
| 5 | 3/4 | 2/4 | 1 | 5/4 |
| 6 | 3/4 | 3/4 | 1 3/4 | 2 |
| 7 | 3/4 | 1 | 2 2/4 | 3 |
| 8 | 1 | 1 | 3 2/4 | 4 |

Table 10.8 shows that the cumulative probability for A is less than for B when the returns are three percent and four percent, but it is not less when the returns are five percent and six percent. This means that the cumulative distributions overlap. Figure 10.19 illustrates this intersection. In this case, the SSD rule cannot be applied.

Figure 10.20 depicts the accumulated cumulative probabilities for the two assets. The accumulated cumulative probability distribution for asset B (solid line) never lies below that for asset A (dotted line). Thus, asset A dominates asset B in SSD.

(*Continued*)

**EXAMPLE 10.6** (*Continued*)

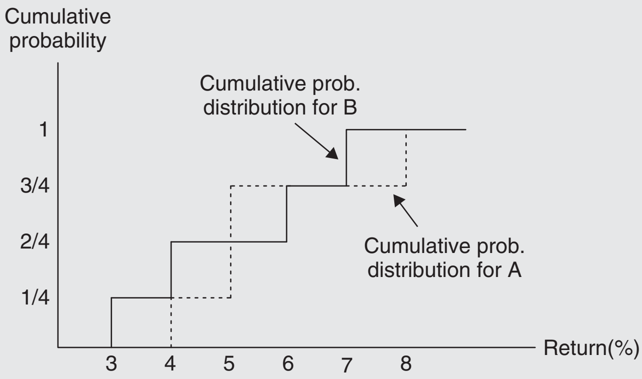


**FIGURE 10.19** Cumulative Probability Distributions for Two Assets

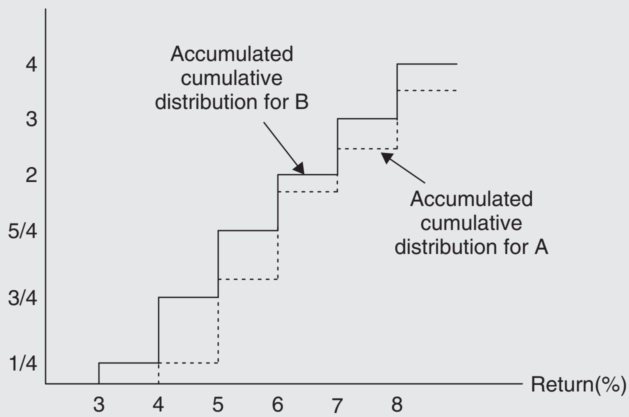


**FIGURE 10.20** Accumulated Cumulative Probability Distributions for Two Assets

### 10.3.3 Third-Order Stochastic Dominance

Any preference between two assets cannot be determined in SSD if the accumulations of the cumulative distributions for the two assets overlap. That is, if $\int_{-\infty}^{r} F_A(x)\,dx \leq \int_{-\infty}^{r} F_B(x)\,dx$ for some value of $r$, but $\int_{-\infty}^{r} F_A(x)\,dx > \int_{-\infty}^{r} F_B(x)\,dx$ for another value of $r$, then neither A nor B can be eliminated by SSD. To determine preferences between these assets by third-order stochastic dominance (TSD), it is necessary to assume that investors exhibit deceasing absolute risk aversion.

*TSD rule: Under the conditions that*

1. investors prefer more to less (equivalently, $U'(r) > 0$),
2. investors are risk averse (equivalently, $U''(r) < 0$),

3. investors exhibit deceasing absolute risk aversion (equivalently, $U^{'''}(r) > 0$), and
4. the expected return of asset A is greater than or equal to that of asset B [i.e., $E_A(r) \geq E_B(r)$],

   asset A dominates asset B in TSD if and only if

   $\int_a^r \int_a^u F_A(x)\,dx\,du \leq \int_a^r \int_a^u F_B(x)\,dx\,du$, or equivalently,

   $\int_a^r \int_a^u [F_A(x) - F_B(x)]\,dx\,du \leq 0$ for all $r$, with the strict inequality holding for at least one value of $r \in [a, b]$, where $a$ and $b$ are the lower and upper bounds for $r$.

TSD implies that the expected utility from asset A is greater than or equal to that from asset B (i.e., $\mathrm{E}[U_A(r)] \geq \mathrm{E}[U_B(r)]$) for all utility functions satisfying the conditions $U'(r) > 0$, $U''(r) < 0$, and $U'''(r) > 0$. See the end-of-chapter appendix for proof. The TSD rule can be applied when the accumulated cumulative probability distributions overlap.

### 10.3.4 Summary of Stochastic Dominance Criterion

The stochastic dominance criteria consist of three forms, depending on the assumptions made regarding the investor's utility function. The FSD criterion requires the assumption $(\partial U/\partial r) > 0$. The SSD criterion requires the assumptions $(\partial U/\partial r) > 0$ and $\left(\partial^2 U/\partial r^2\right) < 0$. And the TSD criterion requires the assumptions $(\partial U/\partial r) > 0$, $\left(\partial^2 U/\partial r^2\right) < 0$, and $\left(\partial^3 U/\partial r^3\right) < 0$. Hence, the TSD efficient set is smaller than the SSD efficient set, which is smaller than the FSD efficient set. In other words, dominance by FSD implies dominance by SSD, and dominance by SSD implies dominance by TSD.

The advantages of selecting investments with stochastic dominance criteria instead of the mean-variance criteria are:

- Stochastic dominance orderings do not presume any particular form of probability distribution (namely, the normal distribution).
- The stochastic dominance criteria imply fewer restrictions on the investor's utility function.
- Undesirable portfolios, such as those on the lower section of the efficient frontier, can be eliminated from further consideration.
- The stochastic dominance selection criteria do not waste information about the probability distribution, every data point is considered. (Unfortunately, the fact that every data point needs to be considered is a deterrent in much empirical work).
- Investors with monotonically nondecreasing utility of wealth functions of any form will always obtain more expected utility from a stochastically dominant asset rather than a dominated one, but some Markowitz-efficient portfolios are not utility maximizers.

One shortcoming of the stochastic dominance criteria is that the stochastic dominance criterion involves pairwise comparisons of all alternatives. When there are many alternatives under consideration, the stochastic dominance criterion may not be feasible. Another shortcoming of the stochastic dominance criteria is that it requires complete knowledge of every point in every asset's probability distribution.

This huge data requirement makes the stochastic dominance criteria impractical for much empirical work. Thus, while the theoretical logic of stochastic dominance is compelling, its usefulness is mostly academic.

## SUMMARY AND CONCLUSIONS

This chapter has analyzed several alternatives to the traditional mean-variance portfolio analysis, which assumes normality (sometimes, lognormality) of asset returns. The decision criterion of the traditional mean-variance portfolio analysis is a risk-return trade-off. In contrast, the theories introduced in this chapter do not assume a particular statistical distribution of returns and are not based on one specific risk-return trade-off. The safety-first criterion, semivariance analysis, and stochastic dominance criteria emphasize downside risk, whereas the portfolio analysis with skewness considers upside potential. These non-mean-variance approaches can also be applied when returns are normally distributed. In this case, many of these nontraditional approaches produce optimal portfolios that lie on the mean-variance efficient frontier.

## APPENDIX A: STOCHASTIC DOMINANCE

This Appendix contains several mathematical proofs that were called for in Chapter 10.

### A10.1 Proof for First-Order Stochastic Dominance

The expected utility of each asset's return is defined as

$$E\left[U\left(r_A\right)\right] = \int_a^b U(r)\ dF_A(r) \quad \text{and} \quad E\left[U\left(r_B\right)\right] = \int_a^{\mathrm{b}} U(r)\ dF_B(r), \tag{A10.1}$$

where $F_A(\cdot)$ and $F_B(\cdot)$ are the cumulative distribution of the return of assets A and B, respectively, and the return $r \in [a, b]$. It is said that asset A dominates asset B in FSD if the expected utility of A's return is greater than the expected utility of B's return. That is, asset A dominates asset B in FSD if

$$\int_a^b U(r)\ dF_A(r) > \int_a^b U(r)\ dF_B(r) \tag{A10.2}$$

or equivalently

$$\int_a^b U(r)\ d\left[F_A(r) - F_B(r)\right] > 0 \tag{A10.3}$$

Using the integration by parts,

$$E\left[U\left(r_A\right)\right] - E\left[U\left(r_B\right)\right] = \int_a^b U(r)\ d\left[F_A(r) - F_B(r)\right]$$

$$= U(r)\left[F_A(r) - F_B(r)\right]\|_a^b - \int_a^b U'(r)\left[F_A(r) - F_B(r)\right]\ dr$$

Now, $F_A\left(b\right) = F_B\left(b\right) = 1$ and $F_A(a) = F_B(a) = 0$. Thus,

$$\int_a^b U(r)\ d\left[F_A(r) - F_B(r)\right] = -\int_a^b U'(r)\left[F_A(r) - F_B(r)\right]\ dr \tag{A10.4}$$

If the right-hand term in equation (A10.4) is positive, asset A dominates asset B in FSD. Because $U'(r) > 0$, $\int_a^b U(r)\ d\left[F_A(r) - F_B(r)\right] > 0$, or, equivalently, $E\left[U\left(r_A\right)\right] - E\left[U\left(r_B\right)\right] > 0$, if $F_A(r) \leq F_B(r)$ for all $r$ with the strict inequality holding for at least one value of $r$.

## A10.2 Proof That $F_A(r) \leq F_B(r)$ Is Equivalent to $E_A(r) \geq E_B(r)$ for Positive $r$

Replacing the return $r$ with $1+r$ in the preceding equations does not affect the argument. Without loss of generality, thus, we still use the restriction that $r$ is positive.

For positive values of a random variable $r$, the expected return is defined as

$$E(r) = \lim_{n\to\infty}\int_0^n r\ dF(r) \tag{A10.5}$$

Using integration by parts,

$$\int_0^n r\ dF(r) = r\ F(r)\ |_0^n - \int_0^n F(r)\ dr$$

$$= n\ F(n) - \int_0^n F(r)\ dr$$

$$= -n\ \left[1 - F(n)\right] + \int_0^n \left[1 - F(r)\right] dr \tag{A10.6}$$

However, $n\ \left[1 - F(n)\right] = n\int_n^\infty f(r)\ dr < \int_n^\infty r\ f(r)\ d = 0$ as $n \to \infty$. Thus, $\lim\limits_{n\to\infty} n\ \left[1 - F(n)\right] = 0$. From equations (A10.5) and (A10.6),

$$E(r) = \int_0^\infty \left[1 - F(r)\right] dr \tag{A10.7}$$

Therefore, if $F_A(r) \leq F_B(r)$, then $E_A(r) \geq E_B(r)$ for positive $r$. Figure A10.1 graphically shows the expected value of the return. The shaded portion is the area

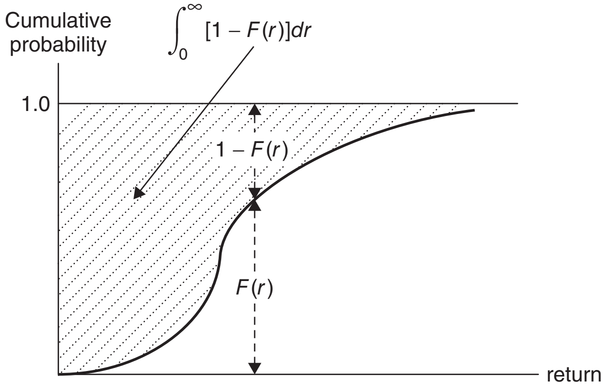


**FIGURE A10.1** Expected Value of a Random Variable and its Cumulative Distribution

under the curve $1 - F(r)$ and it equals $\int_0^\infty [1 - F(r)]\, dr$. Thus, the shaded area is the expected value of the return.

## A10.3 Proof for Second-Order Stochastic Dominance

For the second-order stochastic dominance, we also need to prove that the expected utility of the return on asset A is greater than or equal to that of the return on asset B under certain conditions. From the proof of the first-order stochastic dominance, we have

$$E\left[U\left(r_A\right)\right] - E\left[U\left(r_B\right)\right] \equiv \int_a^b U(r)\ d\left[F_A(r) - F_B(r)\right] = -\int_a^b U'(r)\left[F_A(r) - F_B(r)\right]\ dr \tag{A10.4}$$

We need to prove that the preceding equation is positive under the SSD criteria. Integration by parts of the right-hand side of equation (A10.4) gives

$$\begin{aligned}
&-\int_a^b U'(r)\left[F_A(r) - F_B(r)\right] dr \\
&= -U'(r)\int_a^r \left[F_A(x) - F_B(x)\right] dx\Big|_a^b + \int_a^b U''(r)\int_a^r \left[F_A(x) - F_B(x)\right] dx\, dr \\
&= -U'\left(b\right)\int_a^b \left[F_A(x) - F_B(x)\right] dx \\
&\quad + \int_a^b U''(r)\int_a^r \left[F_A(x) - F_B(x)\right] dx\, dr
\end{aligned} \tag{A10.8}$$

The first term on the right-hand side in equation (A10.8) becomes

$$\begin{aligned}
\int_a^b \left[F_A(x) - F_B(x)\right] dx &= \int_a^b \left[1 - F_B(x)\right] dx - \int_a^b \left[1 - F_A(x)\right] dx \\
&= E_B(r) - E_A(r)] = 0
\end{aligned}$$

Thus,

$$E\left[U\left(r_A\right)\right] - E\left[U\left(r_B\right)\right] = -\int_a^b U'(r)\left[F_A(r) - F_B(r)\right] dr$$
$$= \int_a^b U''(r) \int_a^r \left[F_A(x) - F_B(x)\right] dx\, dr \qquad \text{(A10.9)}$$

If the inequality $\int_a^r F_A(x)\,dx \le \int_a^r F_B(x)\,dx$ holds, then the right-hand side in equation (A10.9) is positive, because $U''^{(r)} < 0$ by definition. Therefore, $E\left[U\left(r_A\right)\right] \ge E\left[U\left(r_B\right)\right]$. The proof is completed.

## A10.4 Proof for Third-Order Stochastic Dominance

In the proof of the second-order stochastic dominance, we have equation (A10.9). For the third-order stochastic dominance, we need to prove that the right-hand side of equation (A10.9) is positive under the TSD criteria. Integration by parts for the right-hand side of equation (A10.9) gives

$$\int_a^b U''(r) \int_a^r \left[F_A(x) - F_B(x)\right] dx\, dr = U''\left(b\right) \int_a^u \int_a^b \left[F_A(x) - F_B(x)\right] dx\, du$$
$$- \int_a^b U'''(r) \int_a^r \int_a^u \left[F_A(x) - F_B(x)\right] dx\, du\, dr \qquad \text{(A10.10)}$$

Because $E_A(r) \ge E_B(r)$, the term in equation (A10.10), $\int_a^b \left[F_A(x) - F_B(x)\right] dx\, du$, is negative, and $U''\left(b\right) < 0$ by definition. Thus, the first term on the right-hand side is positive. Because $U'''(r) > 0$ by definition, the second term on the right-hand side is also positive if $\int_a^r \int_a^u F_A(x)\, dx\, du \le \int_a^r \int_a^u F_B(x)\, dx\, du$. Therefore, $E\left[U\left(r_A\right)\right] \ge E\left[U\left(r_B\right)\right]$. The proof is completed.

# Notes

1. Markowitz (1959) showed that nonquadratic utility functions can be locally approximated with a quadratic utility function. Thus, he argued that the assumption of quadratic utility function is not critical.
2. Leland (1999) showed that when the assumptions of normality and the mean-variance preferences are violated, the CAPM and its risk measures are invalid; the market portfolio is mean-variance inefficient and the CAPM alpha mismeasures the investment performance.
3. The isoelastic utility function is described in Section 4.8.3 of Chapter 4.
4. A stationary portfolio policy can be defined as one in which the same proportions are invested in each asset at the beginning of every period. In a multiperiod setting, such a policy means that at the beginning of each period the investor simply reweights his or her portfolio to have the original proportions. Mossin (1968) has shown that such a policy is optimal if, in addition to the three conditions listed here, a fourth condition is added. This fourth condition is that the distribution of returns must be stationary.
5. For proof, see Young and Trent (1969).
6. See A. D. Roy, "Safety First and the Holding of Assets," *Econometrica* 20 (1952): 431–449.
7. See S. Kataoka, "A Stochastic Programming Model," *Econometrica* 31 (1963): 181–196.
8. Ding and Zhang (2009) establish a risky asset pricing model based on risky funds that is similar to the CAPM.
9. See L. G. Tesler, "Safety First and Hedging," Rev. Econ. Stud. 23 (January 1955–1956): 1–16.
10. The expected return of a variable uniformly distributed between $a$ and $b$ is $\mu = \ a + b \ /2$, and its variance is $\sigma^2 = (b - a)^2/12$.
11. The cumulative probability distribution for the uniform distribution over the interval $[a, b]$ is Prob $(r_i \leq x) = (x - a) / (b - a)$.